

Compositional Generative Models for Out-of-Distribution Generation

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Abstract

Compositional generative modeling aims to create complex unseen combinations by composing multiple pretrained models. Instead of relying on a single large-scale system, individual expert models can be combined at inference time to enable out-of-distribution generation. This approach avoids the need to retrain a new model for every concept pairing. This capability is critical in low-resource settings where specific data combinations are rare. In clinical imaging, for example, hospitals often have many scans for single diseases but very few examples of rare co-morbidities. A compositional system can bridge this gap by synthesizing plausible rare cases from these separate components to support data augmentation.

However, current compositional methods often fail. These systems struggle with tasks like attribute binding, negation, and combining closely related concepts. A primary reason for this failure is the mathematical operator used to combine conditions, which we define as logical composition. This mathematical approach does not always match the semantically meaningful output produced by a single prompt for a complex scene. This creates a disconnect between the mathematical combination and the intended scene. The result limits image quality and makes synthetic data unreliable for sensitive clinical use.

This study investigates this disconnect and develops a method to reduce it. The research follows three steps. First, it characterizes the failure of logical composition through a new taxonomy of concept pairs and trajectory analysis. Second, it tests whether learning more factorized representations, using a partitioned latent space, can align the mathematical combinations with the target scene. Finally, it demonstrates the utility of this framework in medical imaging. The study evaluates whether rare, co-morbid chest X-rays can be synthesized from individual common and separate disease models. This work aims to make out-of-distribution synthesis more reliable and useful for practical downstream tasks.

Declaration

I, Molefe Reabetsoe-phenyo Motlhabane Molefe, hereby declare the contents of this research proposal to be my own work. This proposal is submitted for the degree of PhD in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

Signature

Date

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Chapter 1

Introduction

Diffusion models [Song and Kingma 2021; Ho *et al.* 2020] have become a leading framework for high-fidelity generative modeling. Systems such as Stable Diffusion [Rombach *et al.* 2022], Nana Banana Pro and DALL-E 3 [Betker *et al.*] have shown that text-conditioned diffusion models can produce realistic images at a level of quality previously difficult to achieve. These principles now extend beyond vision [Rombach *et al.* 2022] to speech [Chen *et al.* 2020], medical image reconstruction [Mei *et al.* 2023; Cao *et al.* 2024], and natural language processing [Austin *et al.* 2021; Li *et al.* 2022; Hoogeboom *et al.* 2021]. As a result, diffusion models are increasingly used as general tools for generating and manipulating digital content.

Their strength comes from their iterative sampling process. Instead of producing an output in a single step, a diffusion model starts from noise and refines it through a sequence of denoising updates. This approach makes generation more stable and controllable than earlier architectures like GANs [Goodfellow *et al.* 2014]. This stability helps explain why diffusion models achieve such high sample quality across many diverse domains.

This progress is especially important in medical imaging [Mei *et al.* 2023; Cao *et al.* 2024]. Diffusion models are now used for image denoising [Ho *et al.* 2020], super-resolution [Saharia *et al.* 2023], and synthetic data generation [Rombach *et al.* 2022]. In low-resource clinical settings, these capabilities are particularly valuable because high-quality data are often scarce [Gröger *et al.* 2025; Nyamawe and Shao 2026]. For example, hospitals may have many images of common single conditions but very few examples of rare co-morbid pathologies. A generative model that can produce anatomically plausible rare cases could therefore support critical downstream tasks such as data augmentation [Holste *et al.* 2024].

One useful way to understand these models is through their connection to energy-based modeling [LeCun *et al.* 2006]. In this view, the score function guides a noisy sample toward regions of high probability. This perspective also suggests a route to compositional generation. If separate models represent different conditions, their scores can be combined at inference time to form a joint generation process. Prior work [Du *et al.* 2023; Du and Kaelbling 2024; Skreta *et al.* 2024; Du *et al.* 2020] has shown that this makes it possible to compose concepts without retraining a new model for every combination.

However, this compositional strategy does not reliably produce the intended result [Huang *et al.* 2025; Bradley *et al.* 2025; Du and Kaelbling 2024]. The “logical composition” defined by the mathematical operator often diverges from the “semantic composition” expressed by a single prompt. In practice, this leads to failures such as incorrect attribute binding and interference between related concepts. In clinical imaging, these failures are especially serious because they can blur distinct pathologies or produce anatomically implausible scans. This gap between logical and semantic composition is the central problem addressed in the proposed study.

Bradley *et al.* [2025] argue that composition cannot be defined purely in terms of the input distributions, and instead requires specifying which aspects of each distribution should be preserved in the composed result. Their “projective composition” perspective suggests that success depends on whether the relevant factors can be separated in a suitable representation, not only on whether scores are added in the right way. This suggests that the key issue is not only how composition is performed, but also how the

underlying representation is organised. If the factors remain entangled, the model does not expose the factorisation needed for successful composition.

This reveals a broader unresolved question in compositional generative modeling. Although there is substantial empirical work on controllable diffusion [Wu *et al.* 2024; Xing *et al.* 2024], the link between compositional success and representational factorisation remains underexplored. Recent analyses [Bradley *et al.* 2025; Du and Kaelbling 2024; Huang *et al.* 2025] suggest that compositional failures often arise because the learned representation does not keep the relevant factors sufficiently separate, so composition produces conflicts or incorrect associations instead of the intended semantic combination. Prior work on disentanglement [Locatello *et al.* 2019; Träuble *et al.* 2021b] shows that latent factor separation is unreliable without suitable inductive bias under correlated data. These observations motivate the hypothesis that failures in compositional diffusion may arise from limitations of the learned representation rather than composition alone.

Accordingly, the proposed study pursues three linked objectives and makes three corresponding contributions:

1. **Characterising the composability gap.** The study first develops a dynamics-based measurement framework for analysing how logical score-based composition diverges from semantic composition across different composition regimes. This yields a systematic account of when the gap emerges and identifies the scenarios in which composition produces consistent failures.
2. **Learning representations for composition.** The study then introduces a novel weakly supervised partitioned VAE architecture designed to disentangle the latent space into factors that remain composable under downstream diffusion-based generation. This provides a controlled test of whether improved representational factorisation can reduce the composability gap.
3. **Demonstrating clinical utility for Out-Of-Distribution generation.** The study finally applies this architecture to the synthesis of rare co-morbid chest X-rays from more abundant single-disease data, establishing a clinically meaningful low-resource use case for compositional diffusion. To our knowledge, this is the first study to investigate composable diffusion models for out-of-distribution generation in medical imaging.

These objectives are operationalised through the three research phases summarised in Table 1.1.

Table 1.1: Research phases, objectives, and core assumptions.

Phase	Objective	Assumptions
Phase 1	Characterise the composability gap by comparing logical and semantic composition across controlled concept-pair regimes.	1. Bayes-style score composition succeeds only when the model already contains a usable approximation to the desired joint configuration. 2. The severity of the composability gap varies systematically with the structure of the underlying concept pair.
Phase 2	Test whether more factorised latent representations reduce the composability gap in a controlled synthetic setting.	1. Known generative factors in dSprites and correlated dSprites provide a valid controlled test bed for held-out composition. 2. If representation quality limits composition, more factorised latent spaces should improve held-out composition under a fixed downstream evaluator.
Phase 3	Test whether partitioned clinical representations support plausible low-resource co-morbid chest X-ray generation.	1. Cardiomegaly and pleural thickening are approximately separable in practice. 2. Shared anatomy and disease-specific effects can be factorised. 3. Disease conditionals are approximately independent given the shared anatomy and generative state.

The remainder of the proposal is organised as follows. Chapter 2 reviews the technical background on VAEs, disentangled representation learning, diffusion models, and compositional diffusion. Chapter 3 situates the proposal within prior work on compositional generation and medical image synthesis. Chapter 4 states the research goals and hypotheses. Chapter 5 presents the three-phase research method used to test them. Chapter 6 outlines the project timeline, success criteria, and expected challenges. Chapter 7 summarises the proposal and its expected contribution.

Chapter 2

Background

This chapter provides the background for the three phases of the study. Section 2.1 introduces variational autoencoders and disentangled representation learning, which motivates Phase 2. Section 2.2 introduces diffusion models, their score-based formulation, and compositional diffusion, which underpin all three phases.

2.1 Variational Autoencoders

A standard autoencoder can learn a compact representation for reconstructing an input, but this representation is not necessarily suitable for generation. A variational autoencoder (VAE) [Kingma and Welling 2013] addresses this limitation by making the latent representation probabilistic and regularising it toward a simple reference distribution. The result is a model that can both reconstruct observed examples and generate new ones from sampled latent codes.

During training, the encoder takes an input image x and produces an approximate posterior distribution $q_\phi(z | x)$ over latent codes where ϕ denotes the encoder’s learnable parameters. In the standard Gaussian case, this distribution is summarised by a mean $\mu_\phi(x)$ and a diagonal covariance $\text{diag}(\sigma_\phi^2(x))$. A latent sample z drawn from this distribution is then passed to the decoder to reconstruct the input. At inference time, we sample $z \sim p(z) = \mathcal{N}(0, I)$ from the prior and decode it directly.

More formally, a VAE defines a latent-variable generative model $p_\theta(x) = \int p_\theta(x | z)p(z)dz$, where θ denotes the learnable parameters of a model intended to approximate the data distribution, $p(z) = \mathcal{N}(0, I)$ is a standard Gaussian prior, and $p_\theta(x | z)$ is a learned decoder. Because the marginal likelihood is intractable, training maximises the evidence lower bound (ELBO) such that the objective function becomes:

$$\mathcal{L}_{\text{VAE}}(\theta, \phi; x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x | z)]}_{\text{reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z | x) \| p(z))}_{\text{regularisation}}, \quad (2.1)$$

where $q_\phi(z | x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x)))$ is the encoder’s approximate posterior. The first term rewards reconstructions that explain the observed data and the second keeps the posterior close to the prior.

2.1.1 Disentangled Representations

A representation is *disentangled* if each latent dimension encodes at most one ground-truth factor of variation, and changing one dimension changes the corresponding attribute without affecting others [Wang *et al.* 2024; Träuble *et al.* 2021a]. Here, a factor means a ground-truth attribute of the data, such as orientation or scale. In a VAE, disentanglement is desirable because it gives the latent space a separable structure, with different latent dimensions capturing different underlying factors of variation rather than mixing them together. This makes the representation easier to interpret and enables controlled changes in generated data, since one attribute can be varied without unintentionally affecting others.

One common way to encourage this structure is to modify the VAE objective. By scaling the KL term by $\beta > 1$ in Equation 2.1, β -VAE [Higgins *et al.* 2016] restricts how much information can pass through the model. This pressure encourages the encoder to organise data into a set of distinct, unrelated

traits. FactorVAE [Kim and Mnih 2018] pursues the same goal through a different modification of the objective, replacing the β multiplier with a Total Correlation penalty $\gamma \text{TC}(q(z))$. This term, $\text{TC}(q(z)) = D_{\text{KL}}(q(z) \parallel \prod_j q(z_j))$, specifically measures how much the different features overlap or depend on one another, pushing them to be more independent. Both methods try to force independence within a single shared code $z \in \mathbb{R}^d$. This works when the data’s features are independent, but if the training data contains hidden patterns or correlations, the model will use them as a shortcut. In those cases, simply adding more regularization is not enough to prevent the model from following correlations in the data [Locatello *et al.* 2019]. Locatello *et al.* [2019] show that unsupervised disentanglement is unreliable without inductive bias, that is, assumptions in the model or training procedure that favour certain latent structures over others. Without such constraints, different random seeds and hyperparameter settings can lead to inconsistent results, and models that appear well disentangled cannot be selected or reproduced reliably without some form of supervision. Moreover, for the datasets they study, higher disentanglement does not consistently improve downstream sample efficiency.

2.1.2 Weakly Supervised Disentanglement

To address this limitation, inductive bias is introduced in two main ways: weak supervision through factor labels [Locatello *et al.* 2020; Träuble *et al.* 2021a] and structural partitioning of the latent space [Louiset *et al.* 2024]. Both approaches aim to make factors easier to separate under correlated training data, where independence-based objectives alone are often insufficient

Weak supervision through training-time predictor heads. The first approach uses training-only predictor heads attached to the encoder representation, as in weakly supervised disentanglement [Locatello *et al.* 2020] and subsequent work on attribute disentanglement [Scaramuzzino *et al.* 2023]. In this setting, the relevant bias comes from auxiliary supervision, that is, an additional label-based training signal used to shape the representation without being required at inference time. This supervision encourages the encoder to keep selected factors explicitly predictable from the latent code, rather than allowing them to mix arbitrarily with other sources of variation. For each known factor k , a small MLP g_k is applied to the deterministic encoder mean $\mu_\phi(x)$ and trained against the ground-truth label y_k using either a cross-entropy or regression loss, depending on the factor type. These heads are used only during training and are discarded at inference, so the supervision helps organise the representation while leaving the generative model unchanged at inference time. Applying the auxiliary loss to $\mu_\phi(x)$ rather than to a stochastic sample z provides a stable, low-variance training signal and matches the deterministic representation typically used in disentanglement evaluation [Eastwood and Williams 2018; Locatello *et al.* 2019]. Under correlated training data, this form of weak supervision has been shown to recover part of the factor structure that independence-based objectives alone fail to separate [Träuble *et al.* 2021b].

Correct-head supervision and adversarial exclusion. Auxiliary supervision can make a selected factor predictable from a designated latent region, but it does not guarantee that the same factor is absent from the rest of the latent code [Locatello *et al.* 2020; Scaramuzzino *et al.* 2023]. In other words, supervision can place information in the intended region without fully preventing it from leaking into others. To describe this setup clearly, we refer to the predictor attached to the intended latent region as the *correct-head*, and to predictors attached to other regions as *wrong-heads*. The correct-head encourages a factor to be recoverable from the right region, while the wrong-heads are used to detect whether that factor is still readable from regions where it should not appear.

Figure 2.1 shows the core idea behind wrong-head suppression. A standard classification loss on a wrong-head would be counterproductive, because it would train the encoder to make the leaked factor easier to predict. To avoid this, Scaramuzzino *et al.* [2023] use the gradient reversal layer (GRL), originally introduced by Ganin *et al.* [2016] for domain-invariant representation learning. In that setting, the goal is to learn features that support the main task while suppressing information about whether an example comes from the source or target domain. The same mechanism can be used here to suppress factor leakage. The GRL leaves the forward pass unchanged, so the wrong-head still learns to detect residual leakage, but during backpropagation it multiplies the gradient sent to the encoder by $-\lambda$. As a result, the wrong-head becomes better at exposing leaked factor information, while the encoder is trained to remove that information from the wrong region. Together, correct-head supervision and GRL-based wrong-head suppression encourage each factor to appear in its designated latent partition and nowhere else, yielding cleaner factor separation than supervision alone [Scaramuzzino *et al.* 2023].

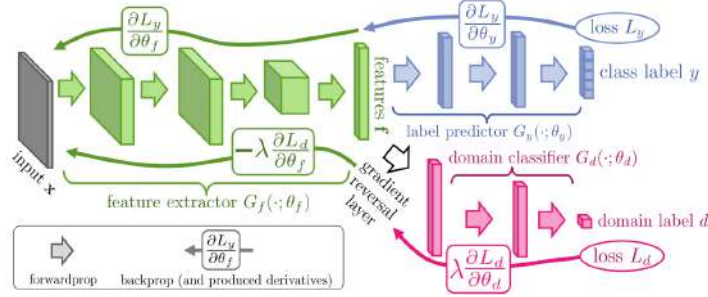


Figure 2.1: Domain-Adversarial Neural Networks (DANN) illustrate the gradient reversal mechanism introduced by Ganin *et al.* [2016]. The architecture describes a label predictor that is trained to solve the main task while a domain classifier is attached through a gradient reversal layer so that the learned features remain uninformative about domain. The same idea can be reused for factor-specific exclusion: a wrong-head tries to detect whether a factor is still present in the wrong latent region, while the reversed gradient pushes the encoder to remove that factor from that region.

Structural latent partitioning across datasets, groups, and modalities. Structural latent partitioning addresses the same problem at the level of model design. Instead of asking a single shared latent code to separate all factors through regularisation alone, these methods assign different sources of variation to different parts of the latent representation from the outset. In practice, this means splitting the latent code into components with different roles, so that shared variation and factor-specific variation are modeled separately from the start. The exact split depends on the structure available in the data, for example whether the setting involves background and target datasets, grouped observations, or multiple aligned modalities.

Contrastive VAE is motivated by settings in which the variation of interest is enriched in a target dataset relative to a background dataset [Abid and Zou 2019]. It partitions the latent space into a common subspace, which captures variation present in both datasets, and a salient subspace, which captures variation specific to the target set. Background samples are reconstructed without using the salient code, which pushes target-specific structure into the salient partition.

Multimodal VAEs are motivated by settings in which the same underlying example is observed through more than one modality, such as paired image and text, or image and segmentation mask [Hsu and Glass 2018; Shi *et al.* 2019]. Their latent spaces are therefore divided into a shared component, which captures information present in all modalities, and private components, which capture variation specific to each modality. This partition is useful for tasks such as joint generation and predicting one modality from another. In practice, however, recovering this separation reliably remains difficult when modality-specific variation dominates [Märtens and Yau 2024].

SepVAE applies the same partitioning logic in a contrastive medical setting [Louiset *et al.* 2024]. Like contrastive VAE, it uses common and salient latent partitions, but it adds constraints designed to reduce leakage between them and to make the salient space more specific to target pathology. In this setting, that means separating normal anatomy shared across cohorts from pathology-specific deviations.

2.2 Diffusion Models

Diffusion models [Sohl-Dickstein *et al.* 2015] are latent-variable generative models of the form $p_\theta(x_0) = \int p_\theta(x_{0:T}) dx_{1:T}$, where $x_{1:T}$ are latents of the same dimensionality as the observed data x_0 . Following Ho *et al.* [2020], the model consists of a fixed *forward diffusion process* that gradually adds noise to data, and a learned *reverse process* that aims to invert this corruption.

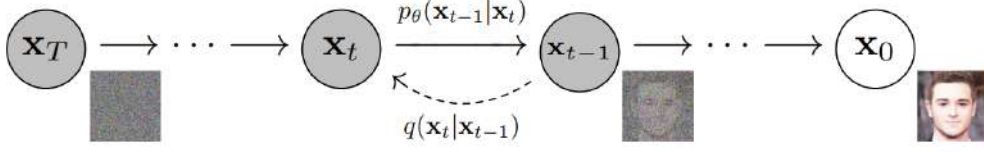


Figure 2.2: Directed graphical model structure of DDPMs [Ho *et al.* 2020].

The joint distribution of all variables is written as a Markov chain:

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t), \quad p(x_T) = \mathcal{N}(0, I), \quad (2.2)$$

where $p_\theta(x_{t-1} | x_t)$ is the reverse process. The forward diffusion process is defined as a separate Markov chain:

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}), \quad q(x_t | x_{t-1}) = \mathcal{N}\left(\sqrt{1 - \beta_t} x_{t-1}, \beta_t I\right), \quad (2.3)$$

where $\{\beta_t\}_{t=1}^T$ is the variance schedule that *can* be learned by reparameterization, or fixed to predetermined constants (e.g., a linear schedule from 10^{-4} to 10^{-2}) such that β_t typically increases slowly with t . This ensures that the forward chain remains analytically tractable and preserves the same functional form across timesteps while gradually driving x_T toward an isotropic Gaussian.

The authors [Ho *et al.* 2020] choose T large (e.g. $T = 1000$) and let β_t increase linearly or according to a smooth curve from a very small value to a modest value. This ensures that each step only slightly degrades the signal, so the forward chain remains close to a diffusion process with well-behaved Gaussian conditionals, and by the final step x_T is nearly standard Gaussian.

Forward Diffusion Process. Because each transition in (2.3) is linear Gaussian, the forward chain admits a closed-form marginal at any timestep and repeatedly composing the Gaussian transitions yields:

$$q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I) \text{ where } \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{s=1}^t \alpha_s. \quad (2.4)$$

We can define x_t through an explicit reparameterization

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \varepsilon_{t-1} \quad (2.5)$$

$$= \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1})} \varepsilon_{t-2} + \sqrt{1 - \alpha_t} \varepsilon_{t-1} \quad (2.6)$$

$$= \dots \quad (2.7)$$

$$= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I). \quad (2.8)$$

where the state at timestep t is a combination of the original data and Gaussian noise, with the signal coefficient $\sqrt{\bar{\alpha}_t}$ gradually shrinking as t increases.

Reverse Denoising Process. The reverse process attempts to invert the forward diffusion by learning Gaussian transitions of the form

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t)). \quad (2.9)$$

Here, θ denotes a single shared neural-network parameterization; the dependence on t is provided through a timestep embedding. Because the forward process $q(x_{1:T} | x_0)$ is fixed and Gaussian, the associated conditional distributions $q(x_{t-1} | x_t, x_0)$ admit closed-form expressions. These conditionals arise directly from standard Gaussian identities: they describe how the forward noising process relates adjacent latent variables when conditioned on the original clean sample x_0 . The DDPM model parameterizes $p_\theta(x_{t-1} | x_t)$ to approximate these analytically derived Gaussian conditionals. It has the Gaussian form

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(\tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I), \quad (2.10)$$

with

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x_0 + \frac{\sqrt{1 - \beta_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t, \quad (2.11)$$

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t. \quad (2.12)$$

The model therefore learns $p_\theta(x_{t-1} | x_t)$ so that its mean matches the analytically derived forward-process posterior. For stability, DDPM fixes the reverse-process variance to one of two experimentally evaluated choices:

$$\sigma_t^2 = \beta_t \quad \text{or} \quad \sigma_t^2 = \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t,$$

as reported by Ho *et al.* [2020]. Both forms preserve the Gaussian functional structure needed for closed-form KL divergences in the ELBO, while yielding similar empirical performance.

Training Objective. Training minimizes the negative log-likelihood $-\log p_\theta(x_0)$. Since $x_{1:T}$ is intractable to marginalize exactly, one optimizes a standard variational lower bound:

$$\mathbb{E}_q[-\log p_\theta(x_0)] \leq \mathcal{L}(\theta) = \mathbb{E}_q \left[-\log \frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \right] = \mathbb{E}_q \left[-\log p(x_T) - \sum_{t=1}^T \log \frac{p_\theta(x_{t-1} | x_t)}{q(x_t | x_{t-1}, x_0)} \right] =: L(\theta), \quad (2.13)$$

This expression can be rewritten as a sum of KL terms [Weng 2021]:

$$\mathcal{L}(\theta) = \mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(x_T | x_0) \| p(x_T))}_{L_T} + \sum_{t=2}^T \underbrace{D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t))}_{L_{t-1}} - \underbrace{\log p_\theta(x_0 | x_1)}_{L_0} \right]. \quad (2.14)$$

The term L_T encourages the forward terminal distribution to match the prior. Each L_{t-1} penalizes discrepancies between the true reverse posterior and the learned reverse conditional at that timestep. The final term L_0 plays the role of a reconstruction likelihood for x_0 . Because all of these terms compare Gaussian distributions with known parameters, they admit closed-form expressions, allowing efficient estimation with low-variance gradients.

Simplified Noise-Prediction Objective. The variational bound in (2.14) has the generic form $L_{\text{VLB}} = L_T + \sum_{t=2}^T L_{t-1} + L_0$, where L_T and L_0 do not depend on the parametrization of the reverse mean. The terms L_{t-1} are KL divergences between Gaussians and therefore can be written in closed form. Assuming an isotropic reverse covariance $\Sigma_\theta(x_t, t) = \sigma_t^2 I$, each such term reduces to a mean-squared error between the true posterior mean $\tilde{\mu}_t$ and the model mean μ_θ :

$$L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 \right] + C_t, \quad (2.15)$$

where C_t does not depend on θ .

Using the closed-form forward posterior $q(x_{t-1} | x_t, x_0) = \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t I)$ and the forward reparameterization $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$, the posterior mean can be expressed as

$$\tilde{\mu}_t(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon \right), \quad (2.16)$$

Since x_t is available as model input, it is convenient to parametrize the reverse mean so that the network predicts the noise directly:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_\theta(x_t, t) \right). \quad (2.17)$$

Substituting $\tilde{\mu}_t$ and μ_θ into (2.15) and simplifying gives

$$L_{t-1} = \mathbb{E}_{x_0, \varepsilon} \left[\frac{(1 - \alpha_t)^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|\varepsilon - \varepsilon_\theta(x_t, t)\|^2 \right] + C_t, \quad (2.18)$$

so each KL term is a weighted mean-squared error between the true noise ε and the predicted noise $\varepsilon_\theta(x_t, t)$.

Dropping the timestep-dependent weight yields the commonly used simplified objective

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, x_0, \varepsilon} [\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2], \quad x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad (2.19)$$

which is equivalent to a reweighted form of the ELBO but is simpler to implement and has been observed to yield improved training stability and sample quality in practice.

Training and Sampling Algorithms. The training and sampling procedures are summarized in Figure ?? . Training draws a data sample x_0 , chooses a timestep t uniformly from $\{1, \dots, T\}$, samples $\varepsilon \sim \mathcal{N}(0, I)$, constructs x_t using the forward marginal (2.4), and takes a gradient step on the objective (2.19). Sampling inverts this process: starting from $x_T \sim \mathcal{N}(0, I)$, the model repeatedly applies the reverse transition implied by (2.17),

$$x_{t-1} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_\theta(x_t, t) \right) + \sigma_t z, \quad z \sim \mathcal{N}(0, I),$$

for $t = T, T-1, \dots, 1$, where σ_t^2 is chosen consistently with the forward variance schedule. This iterative denoising procedure gradually transforms white noise into a structured sample x_0 .

Architecture. The reverse-process mean or noise predictor is implemented using a U-Net architecture with residual blocks, skip connections, and multi-scale feature hierarchies [Ho *et al.* 2020]. The timestep t is encoded by a sinusoidal positional embedding and injected into intermediate layers, allowing the network to adapt its predictions to the current noise level. The network outputs $\varepsilon_\theta(x_t, t)$, which is converted to the reverse mean via Equation (2.17), and the variance is typically fixed as $\Sigma_\theta(x_t, t) = \sigma_t^2 I$ to keep training stable.

2.2.1 Score-Based Formulation

Song *et al.* [Song *et al.* 2021] showed that DDPMs can be understood in continuous time through stochastic differential equations. In this view, discrete-time diffusion models and score-based generative models are two parameterizations of the same underlying denoising process.

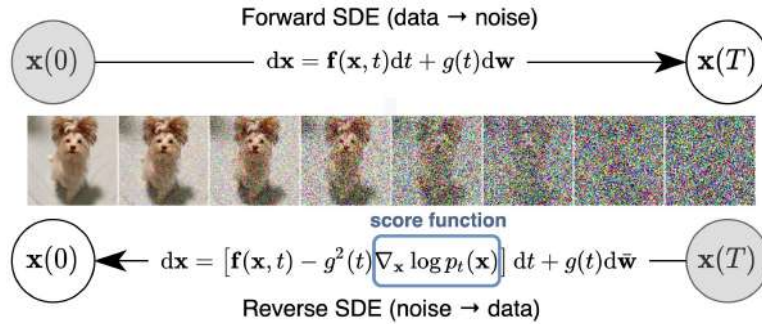


Figure 2.3: **Forward and reverse SDEs.** In continuous time, the forward process corrupts data with drift and diffusion terms, while the reverse process uses the score function to steer noisy samples back toward the data manifold [Song *et al.* 2021].

From the Markov Chain to the Continuous Limit. The discrete forward chain can be viewed as a time discretization of a continuous corruption process. For variance-preserving diffusion, taking the small-step limit yields

$$dx = -\frac{1}{2}\beta(t)x dt + \sqrt{\beta(t)} dW(t), \quad (2.20)$$

where $W(t)$ is a Wiener process. More generally, the forward corruption process is written as

$$dx = f(x, t) dt + g(t) dw, \quad (2.21)$$

with drift $f(x, t)$ and diffusion strength $g(t)$.

Reverse SDE. The corresponding reverse-time SDE has the form

$$dx = [f(x, t) - g(t)^2 \nabla_x \log p_t(x)] dt + g(t) d\bar{w}, \quad (2.22)$$

where $\nabla_x \log p_t(x)$ is the *score function*, namely the gradient of the log-density at noise level t . Intuitively, the score points in the direction that makes the current sample more likely under the time-dependent data distribution. The reverse SDE is the continuous-time counterpart of the DDPM reverse chain where reverse denoising is driven by a vector field that points back toward the data manifold.

In the DDPM parameterization, the model outputs $\varepsilon_\theta(x_t, t)$, an estimate of the Gaussian noise used to corrupt x_0 into x_t . Because x_t is a linear combination of signal and noise, this noise estimate can be converted directly into a score estimate:

$$s_\theta(x_t, t) = -\frac{\varepsilon_\theta(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}}. \quad (2.23)$$

DDPMs can therefore be viewed as score-based generative models written in a different parameterization. The connection is important for Phase 1 because the compositional operators studied there combine score functions from separately conditioned model evaluations, rather than modifying model weights or model architecture.

2.2.2 Conditional Generation and Classifier-Free Guidance

This section distinguishes between **unconditional** and **conditional** diffusion models by examining their objectives, formulations, and applications.

While the above formulation defines the foundation of *unconditional diffusion models*, practical applications often demand controlled or semantically guided generation such as synthesizing specific classes, attributes, or text-aligned imagery. This introduces the need for *conditional diffusion models*, where additional information y (e.g., labels, text embeddings, or medical attributes) guides the reverse process. Recent extensions, such as *classifier* [Dhariwal and Nichol 2021] and *classifier-free guidance* [Ho and Salimans 2022], further refine this conditioning by explicitly or implicitly modulating the score function to balance fidelity and diversity during sampling.

Unconditional Diffusion Models. An *unconditional diffusion model* learns to generate samples from the marginal data distribution $p(x)$ without using auxiliary information or labels. During training, the model observes only samples x_0 drawn from $p_{\text{data}}(x)$ and is optimized to predict the noise added at each diffusion step:

$$\mathcal{L}_{\text{uncond}} = \mathbb{E}_{x_0, t, \varepsilon} [\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2]. \quad (2.24)$$

The training signal is generated from the data itself rather than supplied by external annotation, so the objective is self-supervised. At sampling time, the model starts from Gaussian noise $x_T \sim \mathcal{N}(0, I)$ and denoises it step by step to produce a realistic sample. In this sense, unconditional diffusion models learn $p(x)$ directly, without any conditioning variable.

Conditional Diffusion Models. A *conditional diffusion model* extends the unconditional setting by modeling $p(x | y)$, where y provides additional information such as a class label, text prompt, or spatial map. The forward noising process is unchanged. Conditioning enters only through the reverse model, which is now written as

$$p_\theta(x_{t-1} | x_t, y).$$

In the DDPM parameterization, this is implemented by conditioning the noise predictor $\varepsilon_\theta(x_t, t, y)$. Training therefore uses the objective

$$\mathcal{L}_{\text{cond}} = \mathbb{E}_{x_0, t, \varepsilon, y} [\|\varepsilon - \varepsilon_\theta(x_t, t, y)\|^2]. \quad (2.25)$$

By Equation 2.23, this is equivalent to learning the conditional score of the noisy data distribution. At sampling time, the condition y steers the denoising trajectory toward samples consistent with the desired class, prompt, or attribute. In this way, conditional diffusion turns unconditional generation into controllable generation.

Classifier Guidance. Classifier guidance [Dhariwal and Nichol 2021] provides a post-hoc way to steer an unconditional diffusion model toward a target condition y . The key observation is that a classifier gives access to $p_\phi(y | x_t)$, and therefore to the gradient

$$\nabla_{x_t} \log p_\phi(y | x_t).$$

Applying Bayes’ rule gives

$$p_t(x_t | y) = \frac{p_\phi(y | x_t) p_t(x_t)}{p(y)},$$

so

$$\log p_t(x_t | y) = \log p_\phi(y | x_t) + \log p_t(x_t) - \log p(y),$$

and differentiation with respect to x_t yields

$$\nabla_{x_t} \log p_t(x_t | y) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log p_\phi(y | x_t), \quad (2.26)$$

since $p(y)$ does not depend on x_t . The conditional score can therefore be written as the sum of the unconditional score and a classifier-derived conditioning term. In practice, Dhariwal and Nichol [2021] scale this conditioning term by a guidance weight $\gamma \geq 0$:

$$\tilde{s}(x_t, t, y) = s_\theta(x_t, t) + \gamma \nabla_{x_t} \log p_\phi(y | x_t). \quad (2.27)$$

Larger values of γ strengthen adherence to the target condition, often at the cost of reduced sample diversity.

Classifier-Free Guidance. Classifier-free guidance (CFG) [Ho and Salimans 2022] achieves a similar effect without training a separate classifier. A single diffusion model is trained with condition dropout so that it learns both the conditional and unconditional predictors

$$\varepsilon_\theta(x_t, t, y) \quad \text{and} \quad \varepsilon_\theta(x_t, t, \emptyset),$$

where $\emptyset$ denotes the null condition. At sampling time, these two predictions are combined as

$$\tilde{\varepsilon}_\theta(x_t, t, y) = (1 + w) \varepsilon_\theta(x_t, t, y) - w \varepsilon_\theta(x_t, t, \emptyset), \quad (2.28)$$

with guidance weight $w \geq 0$. When $w = 0$, this reduces to ordinary conditional generation. Larger values of w strengthen the conditional signal while reducing diversity. This formulation has become the standard conditioning mechanism in modern diffusion models. In the proposed study, it plays two concrete roles. This is relevant throughout the proposal because classifier-free guidance provides the basic conditioning mechanism used for standard prompting and for composition. In Phase 1, it underlies both the single-prompt baseline and the compositional operator of Liu *et al.* [2022]. In Phases 2 and 3, the same conditional prediction framework is used in the controlled synthetic and clinical composition settings.

2.2.3 Compositional Diffusion Models

Single text-to-image diffusion models often fail when a prompt requires several factors to be combined correctly. Recent analyses [Huang *et al.* 2025; Du and Kaelbling 2024; Bradley *et al.* 2025] show consistent errors in attribute binding, spatial relations, and numeracy, even when the image remains visually plausible. These failures arise because the models largely learn correlations in joint image-text data rather than an explicit factorized structure over objects, attributes, and relations [Huang *et al.* 2025]. Compositional generative modeling aims to overcome this limitation by generating new factor combinations in a structured and semantically meaningful way. Earlier work framed this in terms of correctness, coverage, and compositionality [Vedantam *et al.* 2017], and models such as SCAN [Higgins *et al.* 2017] showed the value of controllable latent structure. However, the field still lacks a clear account of what a composition should preserve, why some definitions fail, and what structural conditions make composition possible [Bradley *et al.* 2025; Du and Kaelbling 2024].

The next three definitions review the main formulations of composition in this literature: the *simple product*, *Bayes composition*, and *projective composition*. The first two define composition by combining densities directly, while the third defines it through what the composed result should preserve.

Simple Product

The first definition is the *simple product*. Given two distributions p_1 and p_2 over $\mathbb{R}^n$, it defines the composition as

$$\hat{p}(x) \propto p_1(x)p_2(x).$$

This is a familiar way to combine distributions, but it has an important limitation: the composed distribution cannot be truly out-of-distribution with respect to p_1 or p_2 . If either $p_1(x) = 0$ or $p_2(x) = 0$, then $\hat{p}(x) = 0$ as well. As Bradley *et al.* [2025] show, this becomes a problem when the desired composition lies outside the support of both source distributions. In their CLEVR example, if each source distribution contains only one object, then any image with *both* objects has zero probability under both p_1 and p_2 , so the simple product also assigns it zero probability.

Bayes Composition

The second prior definition is *Bayes composition*. Here the idea is not to multiply arbitrary densities, but to interpret the source models as conditional distributions under a shared data distribution. If c_1 and c_2 denote two conditions of interest, then the desired composition is defined as the conditional distribution

$$\hat{p}(x) := p(x \mid c_1, c_2).$$

Assuming c_1 and c_2 are conditionally independent given x , this gives

$$\hat{p}(x) = p(x \mid c_1, c_2) \propto \frac{p(x \mid c_1)p(x \mid c_2)}{p(x)}. \quad (2.29)$$

This is the Bayes composition used in compositional diffusion work such as [Liu *et al.* 2022; Du *et al.* 2023; Du and Kaelbling 2024; Du *et al.* 2020]. It works best when the desired composition already corresponds to a valid conditional distribution under the training data, that is, when the component concepts have been seen together often enough for the model to treat their conjunction as part of the same underlying distribution. In that setting, each conditional contributes evidence for the target sample, while the unconditional term removes shared evidence that would otherwise be counted twice. This is the setting that motivates the co-occurring concept pairs in Group 1 of Phase 1.

However, Bayes composition is not truly out-of-distribution [Bradley *et al.* 2025]. If the desired conjunction is not supported by the underlying data distribution, then the composition cannot create it simply by combining conditionals. It therefore succeeds when the joint concept is already meaningful under the training distribution, and fails when the desired combination lies outside that support.

To use Bayes composition in diffusion models, we convert the composed distribution into a composed score. Starting from

$$\hat{p}(x \mid c_1, c_2) \propto \frac{p(x \mid c_1)p(x \mid c_2)}{p(x)},$$

take the gradient of the log on both sides. This gives

$$\nabla_x \log \hat{p}(x \mid c_1, c_2) = \nabla_x \log p(x \mid c_1) + \nabla_x \log p(x \mid c_2) - \nabla_x \log p(x). \quad (2.30)$$

The composed score is therefore the sum of the two conditional scores minus the unconditional score.

Earlier, the DDPM score was written in terms of the noise predictor as

$$s_\theta(x_t, t, \cdot) = -\frac{\varepsilon_\theta(x_t, t, \cdot)}{\sqrt{1 - \bar{\alpha}_t}}.$$

This means that the same composition can be implemented directly in terms of noise predictions. Let $\varepsilon_\theta(x_t, t, \emptyset)$ denote the unconditional prediction and $\varepsilon_\theta(x_t, t, c_i)$ the prediction conditioned on concept c_i . For a single condition, classifier-free guidance strengthens the effect of that condition relative to the unconditional background through

$$\varepsilon_\theta(x_t, t, \emptyset) + w_i [\varepsilon_\theta(x_t, t, c_i) - \varepsilon_\theta(x_t, t, \emptyset)].$$

Applying this idea to both concepts gives the practical AND operator used by Liu *et al.* [2022]:

$$\tilde{\varepsilon}_{\text{AND}} = \varepsilon_\theta(x_t, t, \emptyset) + w_1 [\varepsilon_\theta(x_t, t, c_1) - \varepsilon_\theta(x_t, t, \emptyset)] + w_2 [\varepsilon_\theta(x_t, t, c_2) - \varepsilon_\theta(x_t, t, \emptyset)]. \quad (2.31)$$

When $w_1 = w_2 = 1$, this reduces to

$$\varepsilon_\theta(x_t, t, c_1) + \varepsilon_\theta(x_t, t, c_2) - \varepsilon_\theta(x_t, t, \emptyset),$$

which is exactly the Bayes composition written in noise-prediction form. This is the practical score-space composition rule used later in Phase 1 and Phase 3.

Projective Composition

Bradley *et al.* [2025] argue that a useful definition of composition must answer two separate questions: what the composed distribution should preserve, and when a practical compositional method can realize it. Their notion of *projective composition* addresses the first question. Under this view, the composed distribution is judged by the part each source is meant to contribute. For example, if one source determines style and another content, then a composition is correct if it preserves the style of the first and the content of the second, even when neither source distribution contains that full combination. This is closely aligned with Du and Kaelbling [2024], where composition is understood as building a new distribution from simpler components that each contribute part of the final structure.

This matters because it allows composition that is out of distribution with respect to the source distributions being combined. The final sample can combine properties from different sources without needing to belong to any one source distribution as a whole. As Figure 2.4 illustrates, the composed distribution can differ from both source distributions in the original sample space while still matching each source after applying the corresponding projection. That is precisely what the simple product cannot express, since it is confined to overlapping support, and what Bayes composition cannot justify when the desired conjunction is not already supported by some underlying joint distribution.

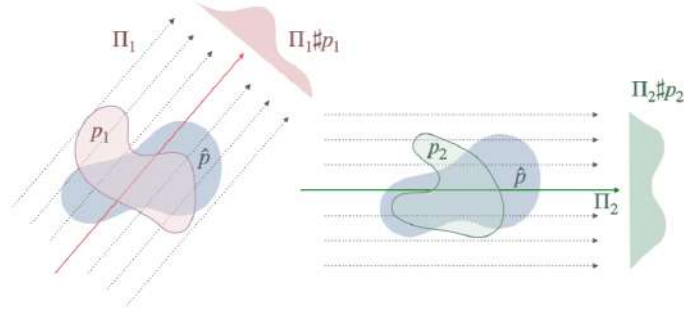


Figure 2.4: Illustration of projective composition. The composed distribution $\hat{p}$ need not coincide with either source distribution p_1 or p_2 in the original sample space. Instead, correctness is defined after applying source-specific projections: under Π_1 , the composed distribution matches the aspect preserved from p_1 , and under Π_2 , it matches the aspect preserved from p_2 . The figure therefore shows how a composition can be out of distribution with respect to the sources as whole distributions while still preserving the designated contribution of each source.

The key implication is representational. Standard disentanglement is not enough if concepts still interfere when one is changed. What composition requires is a stronger separation in which each concept acts on its own factor without perturbing the others. Bradley *et al.* [2025] further show that this separation need not appear in pixel space itself. If it exists in some invertible feature space, then the same composition operator yields a correct projective composition with respect to those feature-space factors. This is a statement about when the target composition is well defined, not a guarantee that reverse-diffusion sampling will always recover it. For the proposed study, the lesson is that compositional success depends on whether the learned representation allows the concepts to be preserved independently.

Chapter 3

Related Work

3.1 Compositional Generative Models

Early approaches decomposed data into disentangled factors of variation, learning representations in a continuous latent space where each dimension corresponds to a semantic factor [Higgins *et al.* 2017; Vedantam *et al.* 2018]. Such factorized representations allowed interpretable control but were limited by their fixed latent dimensionality, which restricted the ability to dynamically introduce or recombine new factors. Complementary efforts in spatial decomposition treated images as structured collections of independent object slots [van Steenkiste *et al.* 2019; Greff *et al.* 2019]. These methods excelled in representing multi-object scenes but often failed to model the interactions or dependencies between objects, which are crucial for understanding relational structure. Despite these limitations, these approaches established the foundation for compositional reasoning in generative models demonstrating that meaningful control arises when latent representations align with distinct conceptual factors.

Energy-Based Models (EBMs) introduced a unifying probabilistic formulation for compositionality. Rather than relying on explicit latent variables, EBMs define an energy function $E_\theta(\mathbf{x})$ that maps an input $\mathbf{x}$ to a scalar energy value, with lower energies corresponding to more likely configurations. Following the product-of-experts formulation introduced by Hinton [1999], independent energy functions can be combined to represent conjunctions of concepts:

$$p(\mathbf{x} \mid c_1, c_2, \dots, c_n) \propto \exp\left(-\sum_i E_\theta(\mathbf{x} \mid c_i)\right).$$

This additive composition provides a mathematically elegant mechanism for representing logical operations such as conjunction (AND), disjunction (OR), and negation (NOT) through energy summation or inversion. Later works such as Shazeer *et al.* [2017]; Greff *et al.* [2019] formalized the notion that multiple energy landscapes can be recursively combined to model increasingly complex visual structures, while Du *et al.* [2020] demonstrated how Markov Chain Monte Carlo (MCMC) sampling on the composed energy surface enables the generation of images consistent with multiple attributes simultaneously.

Building on these principles, Du *et al.* [2020] demonstrated that EBMs can compose concepts probabilistically by defining each semantic concept as an independent energy function. By combining these energy functions, they showed that images satisfying complex logical relationships such as conjunctions, disjunctions, or negations of attributes could be generated. Their work revealed that compositionality need not rely on explicit latent variables or conditional encoders. Instead, it can emerge from the algebra of probability distributions governed by energy functions.

Diffusion Models as Energy-Based Systems. A central insight underpinning recent advances is that diffusion models can be interpreted as a class of energy-based models [Du *et al.* 2023]. The score function $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ of a diffusion model defines a gradient field over data space, which, like an energy landscape, governs the movement of samples toward regions of high likelihood.

Recent work has sought to operationalize compositionality within the framework of diffusion models. Liu *et al.* [2022] propose a structured approach in which separately trained diffusion models, each modeling a distinct component of an image, are algebraically combined at inference time. By viewing diffusion models

as energy-based systems, they define compositional operators, such as logical conjunction (AND) and negation (NOT), that allow for flexible test-time control without additional training. This formulation enables zero-shot compositional generalization, allowing the model to generate coherent images that represent novel concept combinations far beyond its training distribution.

A parallel line of work by Nie *et al.* [2021] introduces latent-space energy-based models for controllable generation. Given a pretrained generator, their approach requires training only a lightweight classifier whose output acts as an energy function over latent codes. This formulation reduces the training complexity of conditional generative models while retaining strong compositional control. Sampling is achieved via ordinary differential equation (ODE) solvers rather than traditional Langevin dynamics [Song and Ermon 2019; Vincent 2011], providing a more efficient and stable inference process.

More recently, Skreta *et al.* [2024] proposed a framework for combining pretrained diffusion models under the principle of *superposition*. Their algorithm, SUPERDIFF, leverages a scalable Itô density estimator for the diffusion stochastic differential equation (SDE), allowing for adaptive reweighting of log-likelihoods during composition. This method enables logical operations such as conjunction and disjunction between pretrained models directly during inference, without retraining or fine-tuning. The authors further demonstrate that the superposition mechanism permits fine-grained temperature control, balancing diversity and coherence in composed samples.

3.1.1 Synthetic Chest X-ray Generation for Medical Imaging

Recent advances in generative modeling have made synthetic image generation an important tool in medical imaging, especially for chest radiography. This is particularly useful where data is scarce, sensitive, or imbalanced across disease classes. Publicly available chest X-ray datasets such as NIH ChestX-ray [Wang *et al.* 2017] and CheXpert [Irvin *et al.* 2019] remain limited by class imbalance, label quality, scale, and privacy constraints. Generative models offer a way to address these limitations by producing anatomically plausible, pathology-aware images that can supplement training data for classification, segmentation, and detection tasks. Synthetic images may also help reduce ethical and legal concerns by replacing identifiable patient data with artificial data that preserve diagnostic relevance [Zhou *et al.* 2025].

Early Generative Approaches Using GANs. Early research on synthetic medical imaging predominantly relied on generative adversarial networks (GANs), which model the data distribution via adversarial training between a generator and discriminator. Frid-Adar *et al.* [2018] and Shin *et al.* [2018] demonstrated that GAN-based augmentation improves convolutional neural network (CNN) performance for lesion classification and anonymization tasks. However, despite their success in general image synthesis, GANs often struggle to maintain fine-grained anatomical and pathological consistency in medical images. The inherent instability of adversarial training and the difficulty of enforcing pixel-level medical realism limit the clinical reliability of GAN-generated data. Consequently, research has shifted toward more stable, probabilistically grounded diffusion-based generative models.

Diffusion Models for Chest X-ray Synthesis. Diffusion models have recently become the dominant approach for high-quality synthetic image generation due to their stability and capacity to model complex data distributions. Mahaulpatha *et al.* [2024] introduced a Denoising Diffusion Probabilistic Model (DDPM) for X-ray synthesis, leveraging iterative noisy-to-clean transformations to reconstruct diverse, realistic radiographs. Unlike GANs, DDPMs explicitly learn the score function of the data distribution, allowing them to generate anatomically consistent samples without adversarial objectives. Their probabilistic nature provides smoother convergence and better control over generation trajectories, making them particularly suitable for medical domains where subtle textural and structural details are critical [Meng *et al.* 2022].

More recent work has transitioned from pixel-space diffusion to latent-space diffusion for improved efficiency and control. Prakash *et al.* [2025] proposed a latent diffusion framework for generating synthetic chest X-rays conditioned on segmentation masks and textual prompts. By guiding the generative process through anatomical or semantic conditions, their method produced pathology-controllable radiographs that enhance the diversity of training data while preserving clinical realism. Their study also integrated radiologist feedback to iteratively refine the generation process as a human-in-the-loop strategy that bridges generative modeling and expert validation. Such approaches directly address the issue of limited publicly available annotated data by enabling targeted synthesis of underrepresented disease cases.

In the context of data privacy, Packhäuser *et al.* [2023] employed a latent diffusion model to generate anonymous, high-quality, class-conditional chest radiographs for thoracic abnormality classification. They argue that biometric identifiers embedded in chest radiographs, such as rib patterns or device signatures, pose re-identification risks that hinder open data sharing. Their proposed model synthesizes clinically valid yet anonymized chest X-rays, thus preserving diagnostic integrity while removing patient-specific identifiers. Comprehensive evaluations demonstrated that classifiers trained on synthetic or mixed (real + synthetic) datasets achieved comparable performance to those trained on purely real datasets, underscoring the feasibility of synthetic data as a privacy-preserving substitute.

Beyond unconditional synthesis, controlling the generation of anatomically and pathologically distinct regions has become a central research focus. Hashmi *et al.* [2024] introduced *XReal*, a pathology- and anatomy-aware diffusion model that introduces spatial control via dedicated anatomy and pathology controllers. By adapting pretrained text-to-image diffusion models, *XReal* enables explicit manipulation of anatomical structures (e.g., lungs, ribs) and pathological regions (e.g., lesions, infiltrates) without requiring full model retraining.

To our knowledge, compositional diffusion has not yet been studied in medical imaging as a method for generating out-of-distribution samples that represent clinically meaningful co-morbid presentations. The proposed study addresses this gap by combining disease-specific diffusion models at inference time to synthesize chest X-rays that reflect multiple pathologies simultaneously, even when such co-morbid cases are scarce or absent in the training data. The contribution is a generative framework for rare co-morbid synthesis in medical imaging.

In summary, diffusion-based generative models represent a significant advancement over prior adversarial methods for medical image synthesis. Their probabilistic and controllable nature enables the generation of realistic, privacy-safe, and pathology-specific chest X-rays that can effectively augment scarce datasets.

Chapter 4

Research Objectives

4.1 Introduction

The previous chapter established the motivation for the proposed research. This chapter defines the research goals and aims of the proposed study.

4.2 Research Goals

The primary research objectives are as follows:

1. **Characterise the Composability Gap.** Measure how severely logical composition diverges from semantic composition, and determine whether this gap varies systematically across concept pairs grouped by co-occurrence support, feature-space disentanglement, or semantic conflict, establishing it as a predictable property of concept pairs rather than a set of isolated failure cases.
2. **Determine the Cause of the Gap.** Identify whether the gap stems from how concepts are represented in the latent space or from the compositional method used at inference time, thereby locating the main bottleneck to reliable compositional generation.
3. **Demonstrate Practical Utility in Medical Imaging.** Test whether compositional generative models can produce plausible rare co-morbid presentations at inference time, without joint training examples, demonstrating that the earlier insights are practically useful in low-resource clinical settings.

4.3 Research Hypotheses

The main hypotheses of this proposal are:

- **H1 (The Composability Gap):** Logical composition via score addition produces a measurable composability gap relative to semantic composition, emerging early in inference and following a principled ordering: smallest for co-occurring pairs, small-to-moderate for approximately disentangled (e.g. style-content pairs) unseen pairs, large for entangled pairs, and largest for semantically competing pairs. The largest-gap outcomes should be unreachable via natural-language prompting alone, confirming that logical composition steers generation along paths no single prompt can reproduce.
- **H2 (The Representation Hypothesis):** The composability gap is primarily a representation problem. Holding the compositional method fixed, a model that keeps concept representations explicitly separate should produce better compositions than one that does not.
- **H3 (The Clinical Independence Sufficiency Hypothesis):** Approximate clinical independence between chest X-ray pathologies is sufficient for logical composition to generate plausible co-morbid outputs. This includes out-of-distribution pairings unseen during training, provided the model has learned representations in which pathology-specific information remains sufficiently distinct.

Chapter 5

Research Method

5.1 Phase 1: Characterising the Composability Gap

As discussed in Section 2.2.3, Bayes-style score composition does not define a truly out-of-distribution joint distribution. Its success therefore depends on the model already containing a usable approximation to the desired joint configuration within its learned training distribution; when that assumption fails, score-level composition can deviate substantially from the behaviour of a model conditioned on a single semantic description of the same scene.

This leads to a central problem:

Logical composition and semantic composition are not equivalent, and their discrepancy depends on the structure of the underlying data distribution.

In this phase, we empirically characterise this discrepancy as a *composability gap*, and study how it varies across different types of concept pairs. To do so, we construct controlled groups of compositions designed to span a range of regimes from approximately factorized pairs to strongly conflicting or entangled concepts. This grouping allows us to probe the structural conditions under which logical composition succeeds or fails, and to identify the mechanisms underlying this gap.

This phase tests Hypothesis 1 in three steps:

- We establish that the composability gap exists and varies systematically across concept pairs chosen to span a range of conditions, from manifold-supported co-occurrence to adversarial concept collision.
- We characterise the gap dynamically by tracking full denoising trajectories, allowing us to identify when the two generation processes diverge and how that divergence develops over time.
- We test whether the endpoint reached by logical composition can be recovered through language conditioning alone. Persistent failure here supports the claim that logical composition reaches targets that standard semantic prompting does not.

5.1.1 Concept Pair Taxonomy

The taxonomy is grounded in the conditions under which Bradley *et al.* [2025] predicts composition to succeed or fail. Rather than treating successes and failures as isolated examples, Phase 1 groups concept pairs into interpretable regimes so that H1 can be tested against a clear structural ordering.

We partition concept pairs into four groups:

1. **Co-occurring Concepts (Group 1):** These pairs frequently co-occur in the training distribution, so the model is likely to have already learned a usable joint representation. This serves as a positive control: logical and semantic composition should remain closely aligned, yielding the smallest gaps.
2. **Feature-Space Disentanglement (Group 2):** These pairs act on largely distinct feature dimensions, such as object–style combinations. They form a second positive control: even without

strong co-occurrence in the training data, composition should remain reliable when the factors are structurally compatible.

3. **Feature Overlap (Group 3):** These pairs have weak joint support and no clear feature-space separation. They are expected to produce intermediate behaviour: larger and more variable gaps than the positive controls, but not necessarily universal failure.
4. **Adversarial Collision (Group 4):** These pairs compete for the same semantic slot, object role, or spatial allocation. They represent the strongest violation of the conditions required for successful composition, and should therefore produce the largest and most systematic failures.

This taxonomy defines the pair regimes used in the analyses below.

5.1.2 Latent Trajectory Dynamics

The purpose of analysing the latent trajectory dynamics is to determine whether the gap between logical and semantic composition appears only at the decoded endpoint or is already present during denoising. To do this, we compare semantic composition (e.g. generating complex through a single prompt) and logical composition under a shared-noise protocol. The analysis will be organised by taxonomy group so that H1 is evaluated with respect to structured compositional regimes rather than a pooled average over heterogeneous pairs.

For each taxonomy group, we select six concept pairs and run twenty-four shared-noise seeds per pair. Each pair-seed record uses the same initial latent state for solo c_1 , solo c_2 , the monolithic prompt, and PoE. Seeds are treated as repeated measurements within a pair, so pair-level summaries remain the unit of group comparison.

This analysis asks three questions: whether the two trajectories remain close or diverge, when that divergence appears during denoising, and whether the expected Group 1–4 ordering is visible in the terminal gap. If the largest and earliest divergences concentrate in the structurally hardest groups, then H1 is supported at the level of latent dynamics rather than only decoded outputs.

This sets up the reachability analysis below.

5.1.3 Reachability Analysis

In this phase, we ask whether the terminal state of logical composition can be reproduced by prompt-conditioned generation. For each pair-seed record, the final composed image is inverted into a continuous prompt text embedding using SD-IPC [Ding *et al.* 2023], and the model is rerun from the same initial noise under that recovered conditioning.

If the regenerated trajectory returns to logical composition’s region, the gap may reflect a limitation of prompt choice within the semantic manifold. If it instead drifts back toward nearby semantic or dominant-concept modes, that supports the stronger claim that logical composition reaches targets that standard prompting does not readily recover. We compare this behaviour across the same Group 1–4 taxonomy, using trajectory closeness, terminal distance to the original PoE run, and output-level similarity as the main indicators.

5.1.4 Summary

Phase 1 characterises the composability gap as a structured empirical phenomenon. The taxonomy identifies the concept regimes in which the gap should be minimal or severe, the trajectory analysis tests whether semantic and logical composition separate during denoising, and the reachability analysis tests whether the endpoint reached by logical composition remains recoverable through prompt-conditioned generation. Taken together, these analyses support H1 when the gap worsens predictably from Groups 1–2 to Group 4, with the largest and earliest divergence appearing in the hardest regimes.

What Phase 1 does not resolve is the cause of this pattern. Because Stable Diffusion [Rombach *et al.* 2022] fixes both the internal representations and the score compositional method, these results cannot yet determine whether the gap is fundamentally a representation problem or a compositional method. Phase 2 addresses that question directly by testing the representation hypothesis in a controlled setting, asking whether our novel architecture that progressively disentangles latents reduces the same compositional failures observed here.

5.2 Phase 2: Explaining the Composability Gap

Phase 1 characterised the composability gap across concept regimes. Phase 2 asks whether that gap is partly a representation problem. The key claim is that if composition is limited by the quality of the learned representation, then more factorised latent spaces should yield more reliable composition under the same downstream conditional diffusion setup.

This distinction is difficult to test directly in a large pretrained model such as Stable Diffusion [Rombach *et al.* 2022], where the learned representation and the inference-time compositional method are tightly coupled. Phase 2 therefore moves to a controlled synthetic setting in which the representation can be varied while the downstream composition pipeline is kept fixed. This leads to a central claim:

If representation quality is a limiting factor for composition, then more factorised latent spaces should yield more reliable compositional generation under the same downstream conditional diffusion architecture.

H2 is supported if better factorised latent spaces improve held-out composition under this shared evaluator.

5.2.1 Dataset Selection

We use dSprites [Matthey *et al.* 2017] and a correlated dSprites variant [Roth *et al.* 2023] because the underlying factors are known and held-out recombination can be measured directly. Standard dSprites serves as the positive control: orientation and scale are sampled independently, and one orientation–scale combination is removed from training. If a model fails to recover that held-out combination in this setting, the limitation is already present before support bias is introduced.

Correlated dSprites: Correlated dSprites keeps the visual domain unchanged but biases orientation–scale co-occurrence. Some combinations become common while others are rare or missing, creating a controlled setting in which the model can rely on shortcut correlations rather than learning the factors separately. This makes it well suited for testing whether better latent factorisation restores held-out composition.

5.2.2 Experiment Design

The experiment compares four regimes under the same downstream evaluation. Regimes A and B establish the baseline effect of changing the training support, Regime C tests whether disentanglement or weak supervision recover performance in the correlated setting, and Regime D tests whether explicit latent partitioning improves further.

Regime A: Control

Regime A trains a baseline VAE on standard dSprites. It asks whether held-out recombination is feasible when the training support is factorised. Success here establishes that the task itself is valid before correlation is introduced.

Regime B: Correlated dSprites Setting

Regime B keeps the same baseline VAE but trains it on correlated dSprites. This isolates the effect of biased support: if performance drops relative to Regime A, then the failure can be attributed to support-induced entanglement rather than a change in model family or evaluation rule.

Regime C: Disentangling Correlated dSprites

Regime C keeps the correlated support of Regime B but strengthens representation learning in two ways. C1 uses standard disentanglement models such as β -VAE and FactorVAE. C2 adds weak supervision on orientation and scale through training-time auxiliary heads while leaving the latent space shared. This regime tests whether stronger independence pressure or limited supervision alone can recover held-out composition under correlation.

Regime D: Partitioned Representation

Regime D keeps the same correlated support and matched supervision as C2, but replaces the shared latent with a partitioned representation. The latent code is organised as $z = (z_c, z_{d1}, z_{d2})$, where z_c stores shared context, z_{d1} orientation, and z_{d2} scale. If Regime D improves beyond C2, the gain is most plausibly due to explicit factor-addressable structure rather than supervision alone.

5.2.3 Evaluation

Phase 2 uses the same evaluation setup in every regime. Generated images are scored with external orientation and scale classifiers trained separately from the representation-learning stage. Each learned latent space is then paired with the same conditional latent diffusion model, so differences in performance can be attributed to the representation rather than the downstream architecture.

Held-out composition test. For each regime, we train the same conditional latent diffusion model in the learned latent space and test composition on held-out orientation–scale combinations. The primary outcome is composition success rate: the proportion of held-out cases in which both target attributes are recovered in the generated image.

Trajectory-level gap closure. As a secondary diagnostic, we compare the composed denoising trajectory with the jointly conditioned trajectory for the same target pair. Smaller terminal gaps indicate that improved representation learning narrows the composability gap rather than only improving decoded outputs.

5.2.4 Summary

Phase 2 addresses the gap identified in Phase 1 by testing H2 in a controlled synthetic setting. Regime A provides the control, Regime B creates the failure setting, Regime C tests whether disentanglement or weak supervision recover performance, and Regime D tests whether explicit latent partitioning improves further. If performance falls from Regime A to Regime B and then improves most clearly in Regime D, Phase 2 supports the claim that representation quality is a meaningful limiting factor for composition.

Phase 3 then asks whether the same idea remains useful for chest X-rays, where conditional independence is approximate and success must be judged by radiological plausibility.

5.3 Phase 3: Compositional Generation of Co-morbid Chest X-rays

This phase is the clinical extension of the representation hypothesis developed in Phases 1 and 2. Phase 1 examines when logical composition fails, and Phase 2 tests whether better-separated representations improve held-out composition in a controlled environment. Building on this, Phase 3 asks whether the same basic idea can be used to generate clinically plausible co-morbid chest X-rays.

We study posteroanterior (PA) chest X-rays containing multiple disease conditions and treat co-morbid synthesis as an out-of-distribution composition problem. The central question is whether the selected diseases are anatomically distinct enough that their effects can be represented as approximately factorized conditionals over shared anatomy. Under this assumption, composition may produce clinically plausible co-morbid images even when joint cases are scarce.

Evaluation focuses on pathology presence, spatial plausibility, radiological fidelity, and held-out co-morbid composition. This phase matters because real co-morbid cases are rare and expensive to annotate. If successful, it would show that compositional generation can help synthesise clinically plausible examples of underrepresented disease combinations.

5.3.1 Dataset Selection

This phase requires a clinical dataset with enough PA chest X-rays, reliable pathology annotations, and meaningful class imbalance to make rare co-morbidity a realistic problem. VinBigData [Nguyen *et al.* 2020] Chest X-ray is well suited to this setting because it provides 15,000 PA radiographs with radiologist

annotations across 14 findings in a strongly long-tailed distribution. Its pathology labels and bounding boxes make it suitable both for selecting a scarce pathology pair and for checking whether disease evidence remains anatomically localised.

As illustrated in Figure 5.1, the dataset is dominated by “No finding” cases, with the remaining pathologies spread across a pronounced long tail. This matters because disease-specific signal can easily be overwhelmed by normal anatomy if the data are used naively. For that reason, training will use balanced sampling across the selected pathologies so that rare findings are not suppressed during learning.

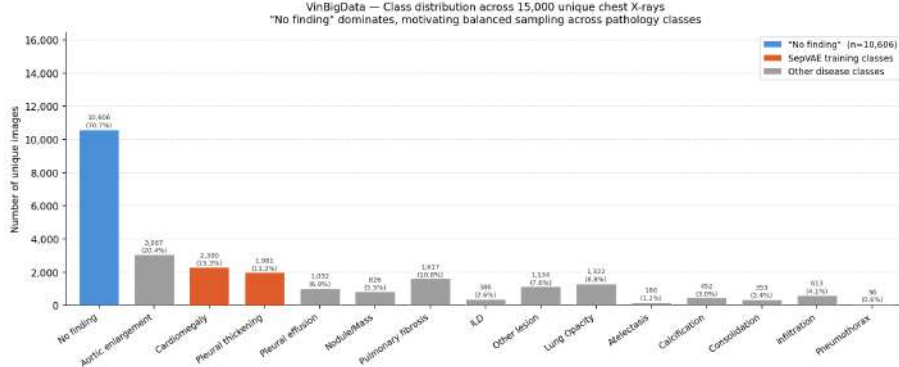


Figure 5.1: Class distribution across 15,000 chest X-rays in VinBigData. The dominance of “No finding” highlights the long-tailed nature of the dataset and motivates balanced sampling during training.

We focus on cardiomegaly and pleural thickening, where cardiomegaly primarily affects the central heart region and pleural thickening appears along the pleural boundary and lung periphery. This pair is clinically plausible because both findings can co-occur in real patients, yet it remains compositionally nontrivial because their joint presentation is relatively rare and both disease effects must be represented within the same underlying chest anatomy. Successful synthesis therefore requires the model to preserve shared anatomical structure while expressing each pathology distinctly, without suppression, leakage, or anatomically implausible blending.

As shown in Figure 5.2, each pathology occurs often enough to support separate learning, but their joint presentation is considerably less frequent: cardiomegaly appears in 2,300 images, pleural thickening in 1,981, and the two are annotated together in only 858. This creates a realistic low-resource co-morbidity setting in which a standard monolithic model sees many examples of each disease in isolation but far fewer examples of how they appear together.

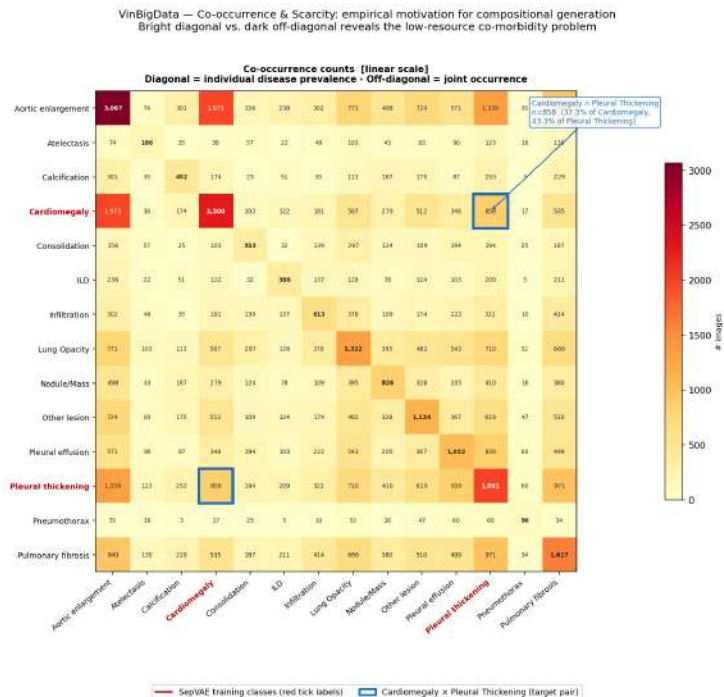


Figure 5.2: VinBigData co-occurrence and scarcity heatmap. The bright diagonal reveals individual disease prevalence, while the darker off-diagonal cells highlight the low-resource co-morbidity problem.

Figure 5.2 therefore defines the low-resource regime in which Phase 3 operates. The model has access to many single-disease examples but comparatively few examples of their joint presentation, so the experiment asks whether the abundant single-disease cases can be used as separate priors and composed to synthesise plausible co-morbid radiographs.

5.3.2 Experiment Design

Preprocessing

Phase 3 begins by matching each raw DICOM radiograph to its pathology labels and bounding-box annotations. Because chest X-rays from different sites can use different intensity conventions, the images are first converted to a common representation before resizing and normalisation. This reduces nuisance variation from acquisition and makes disease-related structure more comparable across patients and clinical sites.

Bounding-box annotations are then retained as weak spatial guidance. They serve two purposes in the proposal. First, they provide an exploratory view of where each selected pathology is typically expressed in the image. As shown in Figure 5.3, cardiomegaly is concentrated around the enlarged cardiac silhouette in the central chest, while pleural thickening appears along the lung periphery and lower pleural margins. Second, they provide weak spatial supervision during training, encouraging disease-related changes to remain within the expected anatomical regions rather than spreading across the chest.

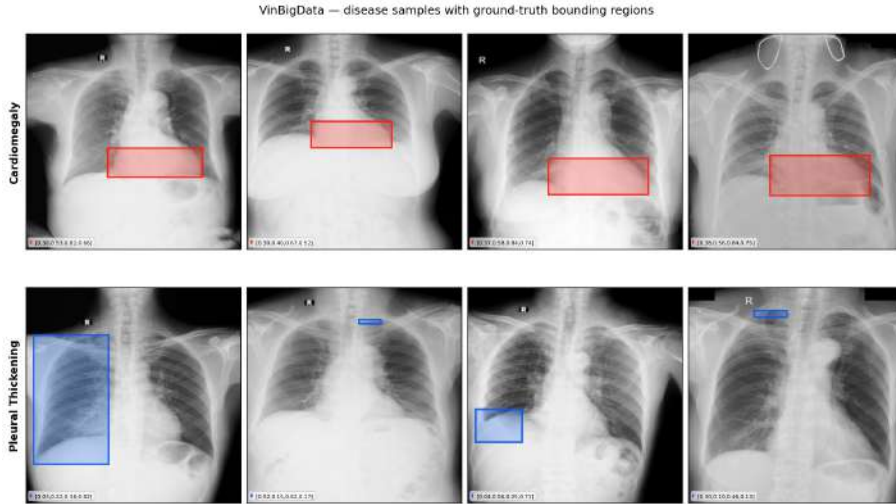


Figure 5.3: Ground-truth examples illustrating the spatial distinctness of Cardiomegaly and Pleural Thickening in VinBigData. Cardiomegaly is concentrated around the cardiac silhouette, whereas pleural thickening is expressed along the lung periphery and pleural margin.

Figure 5.3 does not by itself establish compositional separability, but it shows why this pathology pair is a plausible first test case. The dominant visual evidence for the two diseases tends to occupy different anatomical substrates, which provides a basis for testing whether approximate spatial separation can be turned into a more factorised latent representation.

Conditional Assumptions

As discussed in Section 2.2.3, standard Bayes composition is derived by assuming conditional independence between the composed conditions given the generative variable. In Phase 3, we extend this idea by conditioning on a shared anatomical context z_c and a generative variable u , where u denotes the object over which diffusion operates in this phase. The key modelling step is to assume that, once u and z_c are fixed, the two disease labels do not provide substantial additional information about one another:

$$c_1 \perp c_2 \mid (u, z_c).$$

Under this assumption, the conditional Bayes-style composition becomes

$$p(u \mid z_c, c_1, c_2) \propto \frac{p(u \mid z_c, c_1) p(u \mid z_c, c_2)}{p(u \mid z_c)}. \quad (5.1)$$

In this setting, c_1 and c_2 correspond to cardiomegaly and pleural thickening respectively. This is not a claim that cardiomegaly and pleural thickening occur independently in clinical data. Rather, it is the modelling assumption that once the shared context z_c and generative state u are fixed, the two disease labels can be treated as approximately independent for the purpose of composition. The three assumptions below motivate when this conditional independence is plausible: the selected diseases are approximately separable in practice, shared anatomy can be factorised from disease-specific effects, and under that factorisation the remaining disease conditionals can be treated as approximately independent given (u, z_c) .

1. **The selected diseases are approximately separable in practice.** Cardiomegaly primarily affects the heart region, whereas pleural thickening appears along the pleural boundary and lung periphery. This spatial distinction does not by itself guarantee compositional success, but it provides practical evidence that the two disease effects may be separable in a suitable representation.
2. **Shared anatomy and disease-specific effects can be factorised.** Given this approximate anatomical separability, we assume the representation can be decomposed into shared anatomy and disease-specific components. In particular, common chest anatomy and acquisition variation can be separated from disease-related variation, and the two disease effects can be represented with limited leakage into one another.
3. **These separations induce approximately factorised conditionals over shared anatomy.** If the first two assumptions hold, then the single-disease conditionals can be treated as approximately factorised over a shared anatomical background. Under this assumption, Bayes-style composition becomes a plausible approximation for generating clinically plausible co-morbid images when joint cases are scarce.

Taken together, these assumptions define the domain in which Phase 3 is expected to succeed. They do not claim that co-morbid composition is guaranteed in clinical data; rather, they identify a setting in which Bayes-style composition is theoretically plausible enough to test. The architecture and training objective that follow are designed to encourage a representation consistent with these assumptions, so that held-out co-morbid synthesis becomes an empirical test of H3 rather than an unconstrained application of score addition.

Figure 5.4 shows that, although the two diseases can co-occur clinically, their dominant visual evidence is concentrated in different anatomical regions. Cardiomegaly is centred on the heart region, whereas pleural thickening is concentrated along the peripheral lower pleural margins.

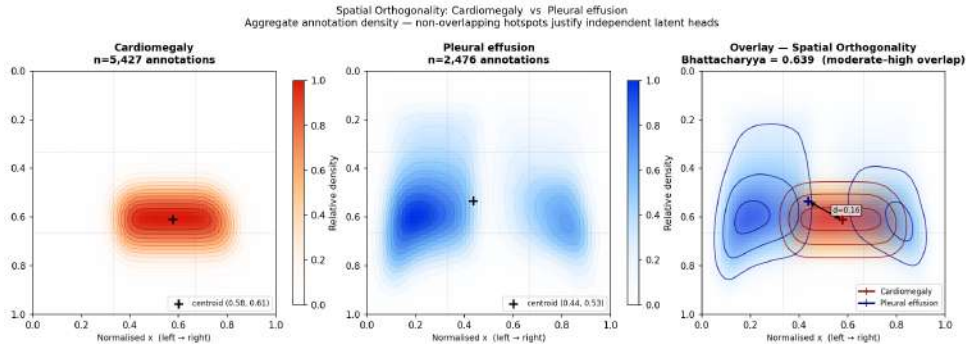


Figure 5.4: Aggregate annotation density for Cardiomegaly ($n = 2,300$) and Pleural Thickening ($n = 1,981$), showing distinct hotspots in the central heart region and along the peripheral lower pleural margins respectively.

Representation Learning

These assumptions motivate an explicit architectural intervention rather than relying on a monolithic latent to disentangle anatomy and pathology implicitly. Phase 3 therefore introduces a partitioned clinical VAE inspired by SepVAE’s common versus salient partitioning, but adapted to a single-dataset

co-morbidity setting. Instead of separating background and target images across two datasets, the model allocates separate latent roles for shared structure and for each disease-specific effect within the same chest X-ray.

This matters because a monolithic latent can still exploit co-occurrence shortcuts and encode both diseases jointly rather than compositionally. Even with supervision, cardiomegaly and pleural thickening could still leak into the same latent dimensions if that improves reconstruction. To address this, Phase 3 introduces a partitioned code,

$$z = (z_c, z_{d1}, z_{d2}), \quad (5.2)$$

where z_c captures shared anatomy, acquisition context, and other non-pathological variation, while z_{d1} and z_{d2} capture the two disease-specific factors. In the present setting, z_{d1} is aligned with cardiomegaly-specific evidence and z_{d2} with pleural-thickening-specific evidence. The encoder therefore produces three posterior heads, one for each latent part, whose samples are concatenated and decoded into the reconstructed radiograph. In the compositional formulation below, z_c provides the shared scaffold, while disease-specific effects are represented so they can be composed conditionally over that scaffold.

This latent partition is complemented by training-only auxiliary predictor heads applied to the encoder means of the disease latents. One classifier is trained to recover cardiomegaly from μ_{d1} and the other to recover pleural thickening from μ_{d2} , aligning each latent branch with its intended pathology. To discourage cross-factor leakage, the same predictors are also applied to the opposite latent branch through gradient reversal layers (GRLs), so that the classifiers detect residual leakage while the encoder is trained to remove it. These auxiliary heads are discarded at test time, leaving a generative model whose composition interface is the partitioned latent representation itself.

At a high level, the objective function is composed of the following:

- **Reconstruction and regularisation:** these preserve faithful reconstruction while keeping the shared and disease-specific latent blocks well behaved.
- **Correct-head supervision:** auxiliary predictors applied to μ_{d1} and μ_{d2} encourage each disease-specific latent to encode its intended pathology, making the partition clinically interpretable rather than merely structural.
- **Wrong-head suppression with GRLs:** auxiliary predictors applied to the opposite latent branch through gradient reversal layers discourage cardiomegaly evidence from leaking into z_{d2} and pleural-thickening evidence from leaking into z_{d1} .
- **Dependence reduction:** orthogonality or mutual-information penalties reduce residual statistical and geometric overlap between z_{d1} and z_{d2} beyond what the classifier-based terms remove.
- **Region-weighted reconstruction:** bounding-box-guided reconstruction promotes locality by forcing disease-related changes to be explained primarily within the annotated pathology regions.

Together, these terms preserve a usable generative model while pushing the learned representation toward factor-addressable disease latents. The intended outcome is that shared anatomy remains in z_c , each disease-specific signal is captured in the appropriate latent part, and the two factors can later be combined with less interference at inference time.

Latent Diffusion Model

The partitioned VAE defines the representation, but the actual generative composition is performed by a single conditional latent diffusion model trained on that latent space. We write u_0 for the clean generative variable modeled by the diffusion process and u_t for its noisy version at time t , with generation initialized from $u_T \sim \mathcal{N}(0, I)$. In this setup, diffusion operates over the generative variable u , while z_c provides the shared anatomical context. Composition is performed over u conditioned on z_c , and the null condition $\emptyset$ corresponds to omitting the disease label while keeping that shared context fixed.

We do not train separate unconditional and conditional generators, because classifier-free guidance assumes a single conditional diffusion model that can be evaluated under different conditions. During training, each u_0 is paired with its observed disease label $y \in \{c_1, c_2\}$, and that label is replaced by the null condition $\emptyset$ with probability p_{uncond} . This gives one diffusion model that learns the conditional branches $\varepsilon_\theta(u_t, t \mid z_c, c_1)$ and $\varepsilon_\theta(u_t, t \mid z_c, c_2)$ together with the branch $\varepsilon_\theta(u_t, t \mid z_c, \emptyset)$. Here, $\emptyset$ means no disease label given z_c , not a literal normal or no-finding class. In the low-resource setting of Phase 3,

joint cardiomegaly and pleural-thickening cases are absent or heavily reduced during generative training, but the model still learns the two single-disease conditionals and the null branch required for conditional composition.

Under the conditional independence assumption above, the corresponding score decomposition is

$$\nabla_u \log p(u \mid z_c, c_1, c_2) = \nabla_u \log p(u \mid z_c, c_1) + \nabla_u \log p(u \mid z_c, c_2) - \nabla_u \log p(u \mid z_c). \quad (5.3)$$

In practice, Phase 3 implements the classifier-free guidance analogue of this conditional score decomposition:

$$\begin{aligned} \tilde{\varepsilon}_{\text{AND}}(u_t, t; z_c) &= \varepsilon_{\theta}(u_t, t \mid z_c, \emptyset) \\ &\quad + w_1 [\varepsilon_{\theta}(u_t, t \mid z_c, c_1) - \varepsilon_{\theta}(u_t, t \mid z_c, \emptyset)] \\ &\quad + w_2 [\varepsilon_{\theta}(u_t, t \mid z_c, c_2) - \varepsilon_{\theta}(u_t, t \mid z_c, \emptyset)]. \end{aligned} \quad (5.4)$$

This ε -based form is the practical CFG-style approximation to the conditional Bayes score decomposition, not an exact probabilistic identity. The resulting denoised latent is then decoded through the VAE decoder to produce a candidate co-morbid radiograph, which is passed to the evaluation procedure below. In this way, Phase 3 combines a partitioned representation with a single conditional latent diffusion model that implements Bayes-style composition over a shared scaffold.

5.3.3 Evaluation

Phase 3 evaluates whether a factorised clinical representation supports held-out co-morbid composition under realistic conditions. As in Phase 2, the goal is to test whether the intended factors remain recoverable after composition. The criteria below are applied to the candidate radiograph produced by conditional composition over the shared scaffold z_c . Final testing is performed in a low-resource setting where joint cases are absent or heavily reduced during generative training while single-disease cases remain available.

Each composed image is assessed along four dimensions:

- **Pathology presence:** two external classifiers, trained on splits disjoint from generative-model training, test whether the composed image contains recoverable evidence for both cardiomegaly and pleural thickening.
- **Co-morbidity fidelity:** a third classifier, trained on real co-morbid chest X-rays, tests whether the image resembles a plausible joint presentation rather than an artificial overlay of isolated findings.
- **Spatial plausibility:** disease evidence should remain aligned with the expected heart-region and pleural-region territories, without substantial cross-region leakage.
- **Held-out composition success:** composition is counted as successful when the generated sample is scored positive by both pathology classifiers, receives a high co-morbidity fidelity score, and satisfies the spatial plausibility check.

5.3.4 Summary

Phase 3 transfers the representation hypothesis of Phase 2 to a clinical low-resource co-morbidity setting using VinBigData. It assumes that shared anatomy can be represented by z_c , disease-specific effects can be factorised from that shared context, and composition can therefore be performed over the generative variable u conditioned on z_c . To test this, Phase 3 combines a partitioned latent architecture, a factor-separating objective, and a single conditional latent diffusion model.

The central test is whether this representation supports held-out co-morbid composition. If composed samples recover both target pathologies, remain spatially plausible, and are judged consistent with real co-morbid radiographs, then Phase 3 supports H3. Otherwise, either the conditional assumptions are too weak in this clinical setting or the learned representation is still not sufficiently factorised to support stable composition.

Chapter 6

Research Plan

6.1 Time Plan

	2024					2025												2026												2027					
	08	09	10	11	12	01	02	03	04	05	06	07	08	09	10	11	12	01	02	03	04	05	06	07	08	09	10	11	12	01	02	03	04	05	06
Paper Reading																																			
Writing Proposal																																			
Phase 1																																			
Phase 2																																			
Phase 3																																			
Write Thesis																																			

6.2 Success Criteria

- Phase 1 should deliver a stable taxonomy benchmark and trajectory-based evidence that the composability gap grows predictably from structurally easy pairs to conflicting pairs.
- Phase 2 should show that better factorised latent representations improve held-out composition and reduce the gap between semantic and logical composition under a shared downstream evaluator.
- Phase 3 should produce VinBigData compositions that preserve both target pathologies, remain spatially plausible, and score as credible co-morbid radiographs under external evaluation.
- The full programme should end with reproducible manifests, evaluation scripts, and thesis-ready chapters that support publication and final submission.

6.3 Possible Challenges

- Diffusion training, shared-noise sweeps, and repeated ablations are computationally expensive and may slow iteration on limited hardware budgets.
- Composition quality can be sensitive to seeds, hyper-parameters, and representation choices, so multiple controlled runs are needed before drawing conclusions.
- Real co-morbid chest X-ray cases remain scarce, which limits the strength of supervision available for the clinical phase.
- Clinical validity depends partly on external classifiers and proxy realism checks, so failure analysis and specialist interpretation remain essential.

6.4 Conclusion

This plan moves from characterising the composability gap, to testing whether better factorised representations reduce it, and finally to evaluating whether the same logic supports clinically plausible co-morbid chest X-ray synthesis. The phased structure keeps each stage tied to concrete outputs, making the overall programme feasible, cumulative, and coherent for thesis completion.

Chapter 7

Conclusion

This proposal argues that the main barrier to reliable compositional generation is the mismatch between logical composition and the semantically correct joint concept. To address this problem, the project proceeds in three stages: first, it characterises the composability gap through a structured benchmark and trajectory-based analysis; second, it tests whether more factorised latent representations can reduce this gap in a controlled setting; and third, it evaluates whether the same principles can support clinically plausible synthesis of rare co-morbid chest X-rays.

Taken together, the proposed work aims to contribute both a clearer theoretical account of why compositional diffusion models fail and a practical pathway for improving out-of-distribution generation in low-resource domains. If successful, the thesis will show that reliable composition depends not only on how conditions are combined at inference time, but also on how cleanly the underlying concepts are represented, with direct implications for generative modeling and medical image synthesis.

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